

Announcements

Classes you might be interested in next semester:

- ✱ ■ CS 3350 Intro to Practical ML ✱
- ✱ ■ CS 5955/6955 Advanced AI
- CS 6370 Motion Planning
- CS 6960 Multimodal LLM Agents

RL + LLMs

Midsemester Feedback

- What is going well:

- Projects are engaging and a good way to see practical benefits of ideas from class.
- Homeworks are a good way to practice theory before implementation.
- Good lecture slides
- Recorded lectures!
- Approachable instructor
- In class quizzes are useful

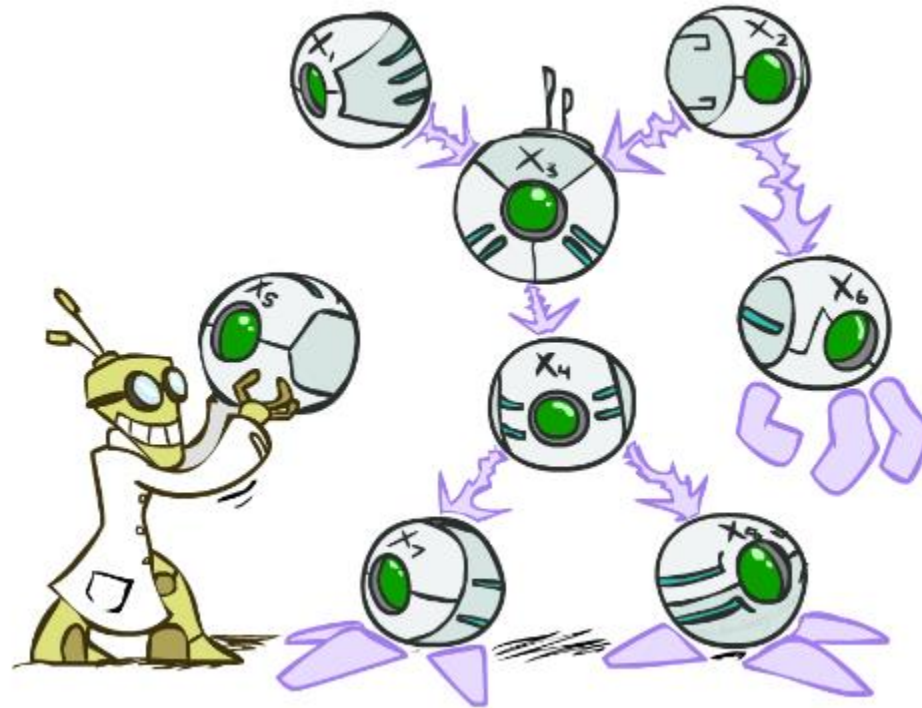


Ways to Improve

- Some topics covered too quickly (e.g. nnets, MDPs)
- More spacing between programming projects and homeworks
- Improve grading speed
- Include lecture annotations on slide pdfs
- More worked examples
- More TA hours, online TA options...
- More in-class activities to break things up
- Use of modern libraries (e.g., PyTorch, Gymnasium)
- More heads up on assignment deadlines.
- More chances for questions and interactivity during lecture

CS 6300: Artificial Intelligence

Bayes' Nets



Instructor: Daniel Brown --- University of Utah

[Based on slides created by Dan Klein and Pieter Abbeel <http://ai.berkeley.edu>.]

Probabilistic Models

- Models describe how (a portion of) the world works
- **Models are always simplifications**
 - May not account for every variable
 - May not account for all interactions between variables
 - “All models are wrong; but some are useful.”
– George E. P. Box
- What do we do with probabilistic models?
 - We (or our agents) need to reason about unknown variables, given evidence
 - Example: explanation (diagnostic reasoning)
 - Example: prediction (causal reasoning)
 - Example: value of information



$$P(x|y,z) = P(x|z)$$

Probability Recap

- Conditional probability

$$P(x|y) = \frac{P(x,y)}{P(y)}$$

Bayes' Rule

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Product rule

$$P(x,y) = P(x|y)P(y)$$

- Chain rule

$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$

- X, Y independent if and only if: $\forall x, y : P(x,y) = P(x)P(y)$

$X \perp\!\!\!\perp Y$

- X and Y are conditionally independent given Z if and only if:

$$X \perp\!\!\!\perp Y | Z$$

$$\forall x, y, z : P(x,y|z) = P(x|z)P(y|z)$$

Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- *Conditional independence* is our most basic and robust form of knowledge about uncertain environments.

- X is conditionally independent of Y given Z

$$X \perp\!\!\!\perp Y | Z$$

if and only if:

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

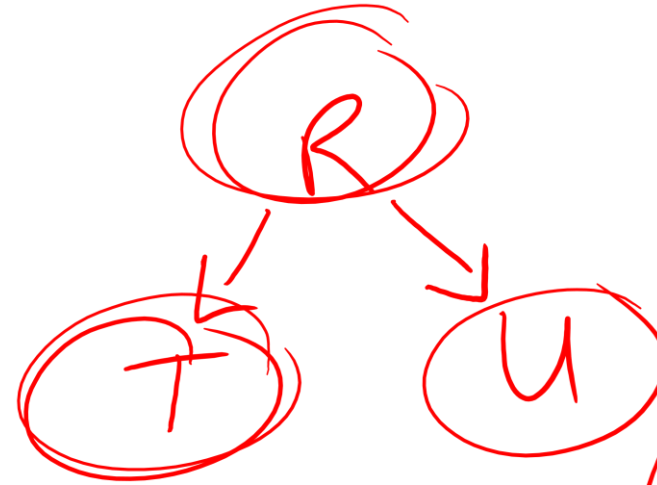
or, equivalently, if and only if

$$\forall x, y, z : P(x | z, y) = P(x | z)$$

Conditional Independence

- What about this domain:

- Traffic
- Umbrella
- Raining



$$U \perp\!\!\!\perp T \mid R$$

$$P(U \mid R) = P(U \mid R, T)$$



Conditional Independence and the Chain Rule

- Chain rule: $P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots P(X_n|X_1, \dots, X_{n-1})$

- Trivial decomposition:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$$

- With assumption of conditional independence of T and U given R:

$$P(\text{Traffic}, \text{Rain}, \text{Umbrella}) = P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$

- Why useful?
- Bayes' nets / graphical models help us express conditional independence assumptions



Ghostbusters Chain Rule

- Two places to check for a ghost (top, bottom).

$$P(G)P(T|G)P(B|G)$$

- Each sensor depends only on where the ghost is
- That means, the two sensors are conditionally independent, given the ghost position

$$P(T,B,G) = P(G) P(T|G) P(B|G)$$

T	B	G	P(T,B,G)
+t	+b	+g	0.16
+t	+b	-g	0.16
+t	-b	+g	0.24
+t	-b	-g	0.04
-t	+b	+g	0.04
-t	+b	-g	0.24
-t	-b	+g	0.06
-t	-b	-g	0.06

- T: Top square is red
- B: Bottom square is red
- G: Ghost is in the top



- Givens:

$$P(+g) = 0.5$$

$$P(-g) = 0.5$$

$$P(+t | +g) = 0.8$$

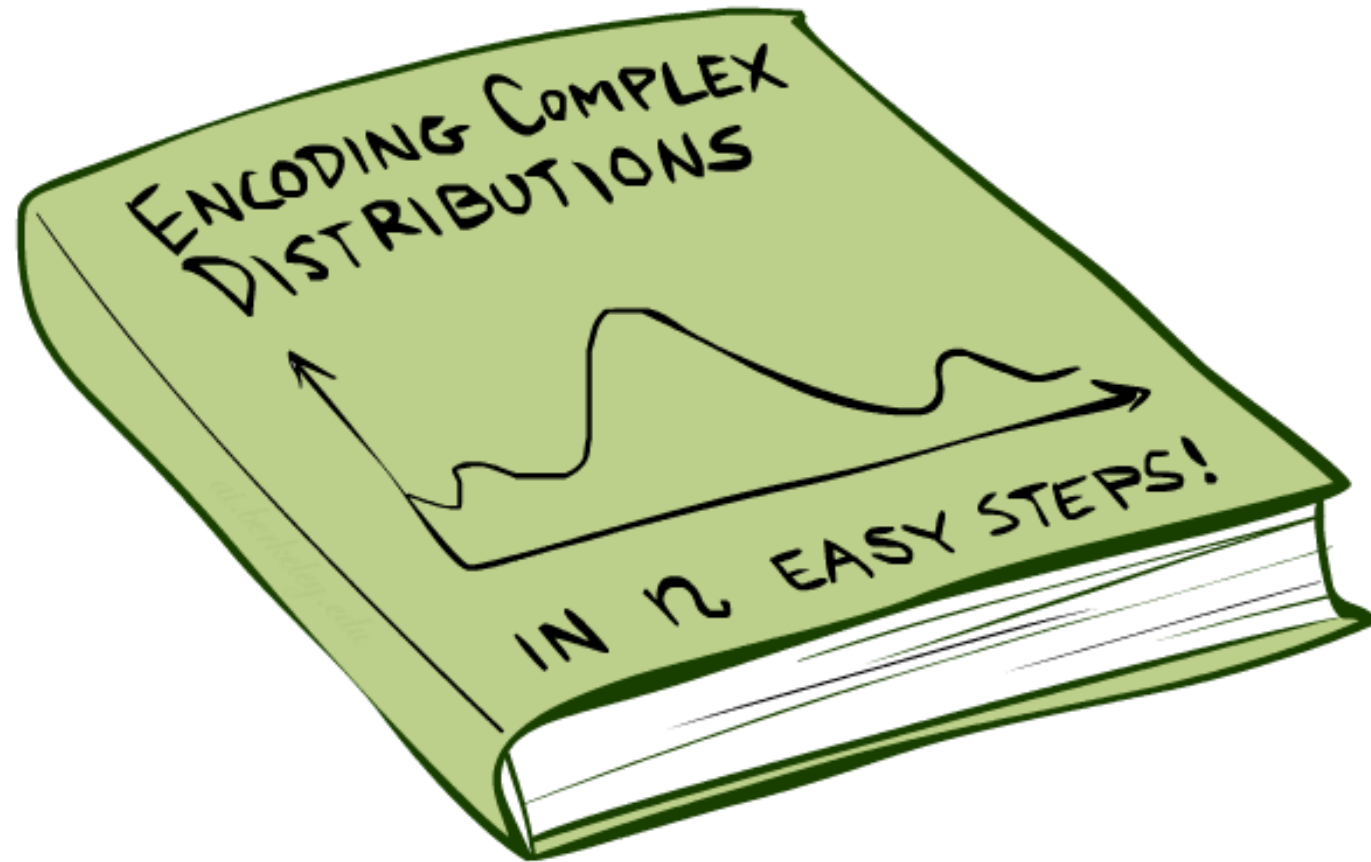
$$P(+t | -g) = 0.4$$

$$P(+b | +g) = 0.4$$

$$P(+b | -g) = 0.8$$

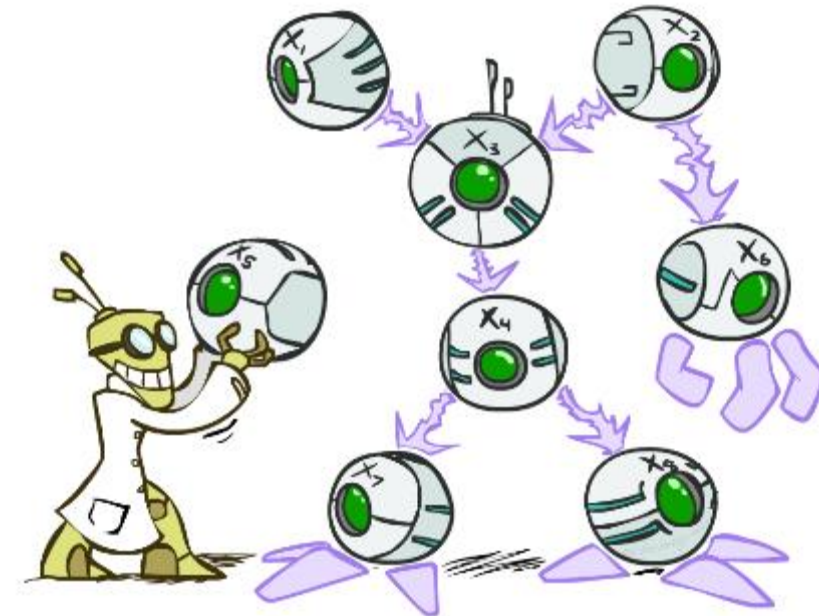


Bayes' Nets: Big Picture

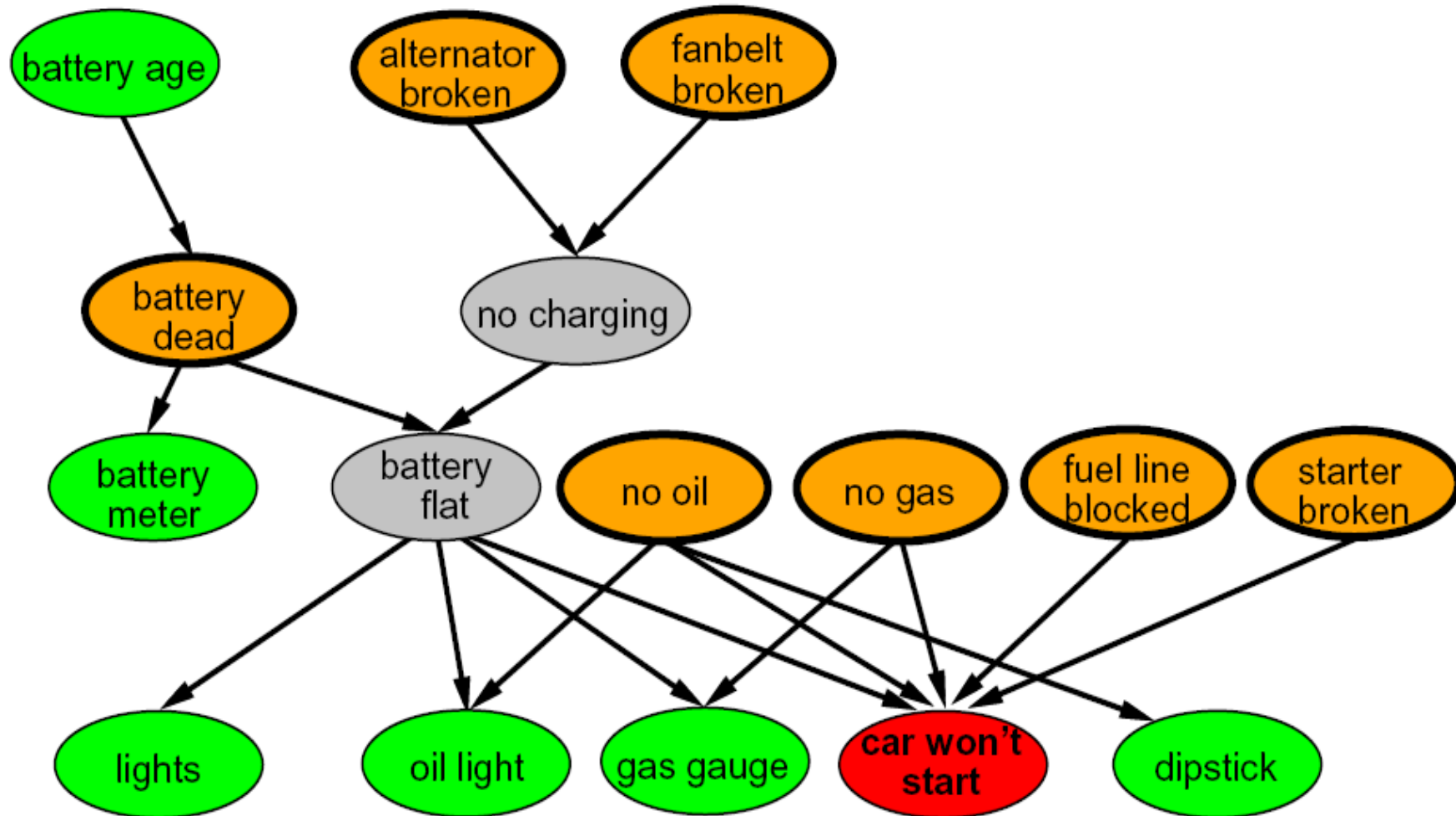


Bayes' Nets: Big Picture

- Two problems with using full joint distribution tables as our probabilistic models:
 - Unless there are only a few variables, the joint is WAY too big to represent explicitly
 - Hard to learn (estimate) anything empirically about more than a few variables at a time
- **Bayes' nets:** a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
 - More properly called **graphical models**
 - We describe how variables locally interact
 - Local interactions chain together to give global, indirect interactions
 - For about 10 min, we'll be vague about how these interactions are specified

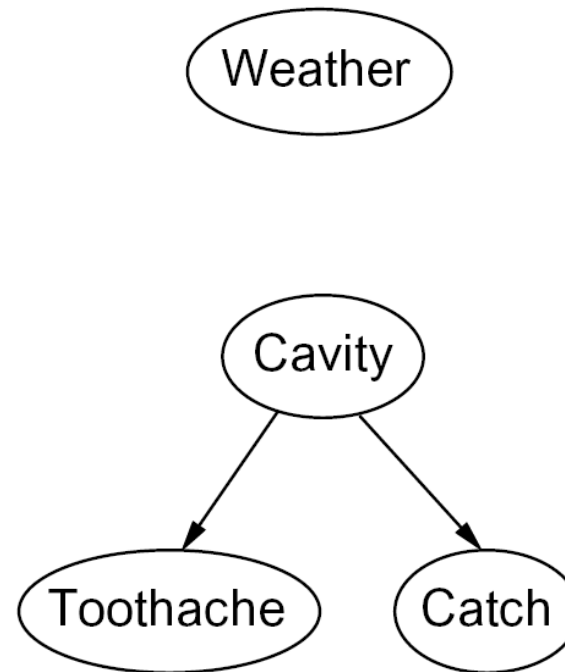


Example Bayes' Net: Car



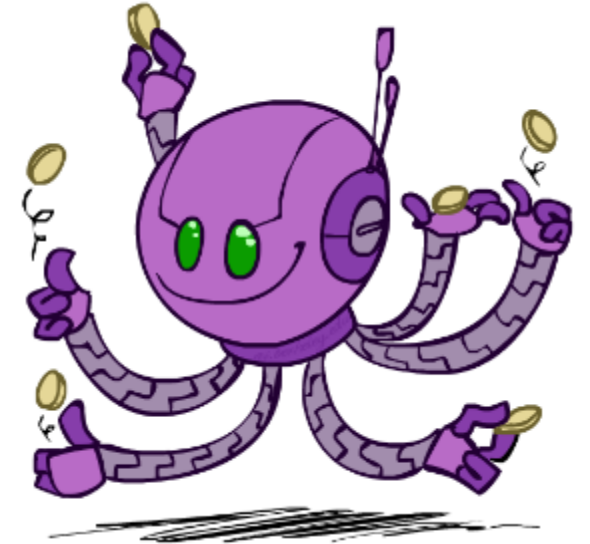
Graphical Model Notation

- **Nodes: variables (with domains)**
 - Can be assigned (observed) or unassigned (unobserved)
- **Arcs: interactions**
 - Indicate “direct influence” between variables
 - Formally: encode conditional independence (more later)
- For now: imagine that arrows mean direct causation (in general, they don't!)



Example: Coin Flips

- N independent coin flips



- No interactions between variables: **absolute independence**

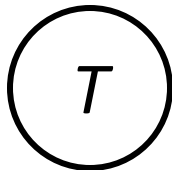
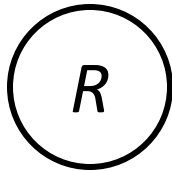
Example: Traffic

- Variables:

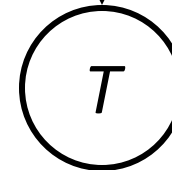
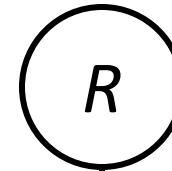
- R: It rains
- T: There is traffic



- Model 1: independence



- Model 2: rain causes traffic



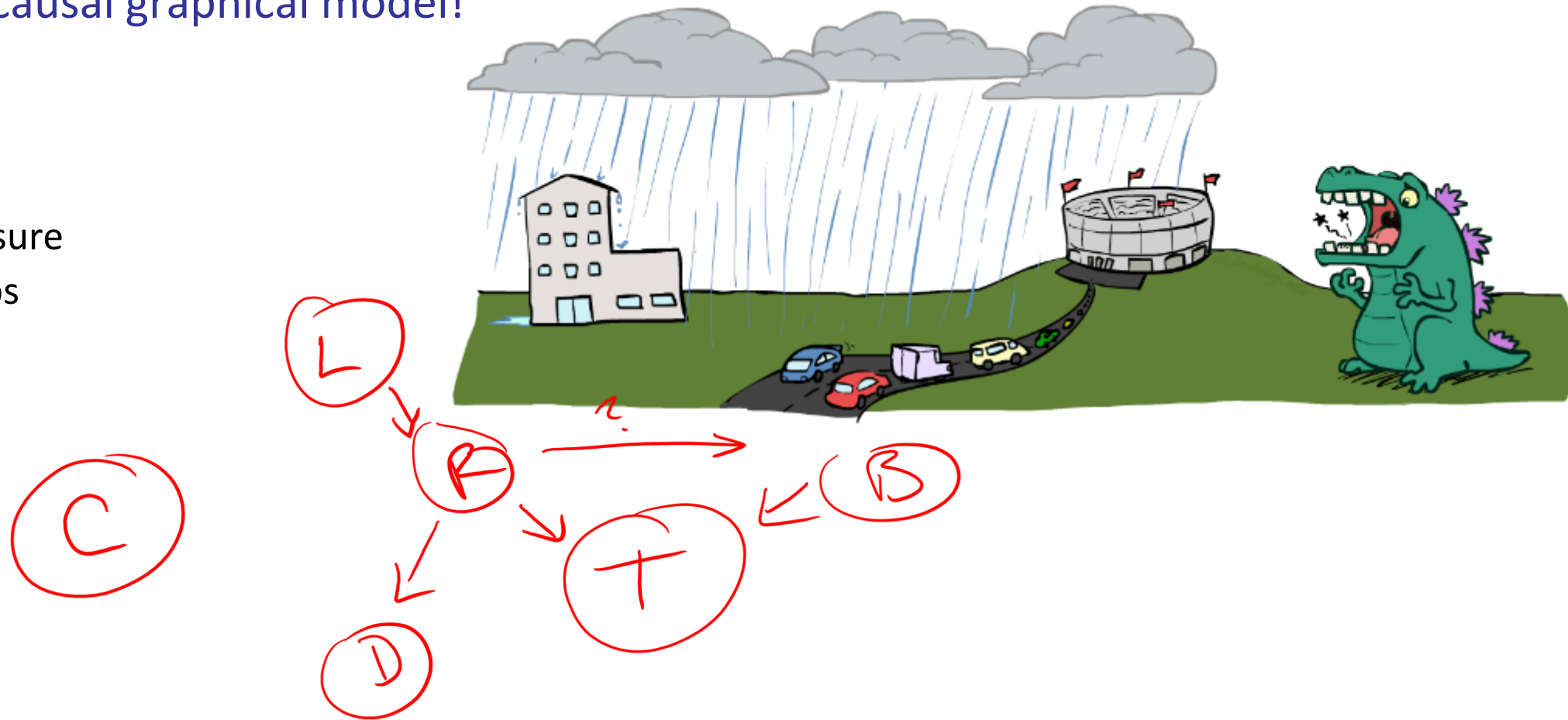
- Why is an agent using model 2 better?

Example: Traffic II

- Let's build a causal graphical model!

- Variables

- T: Traffic
- R: It rains
- L: Low pressure
- D: Roof drips
- B: Ballgame
- C: Cavity

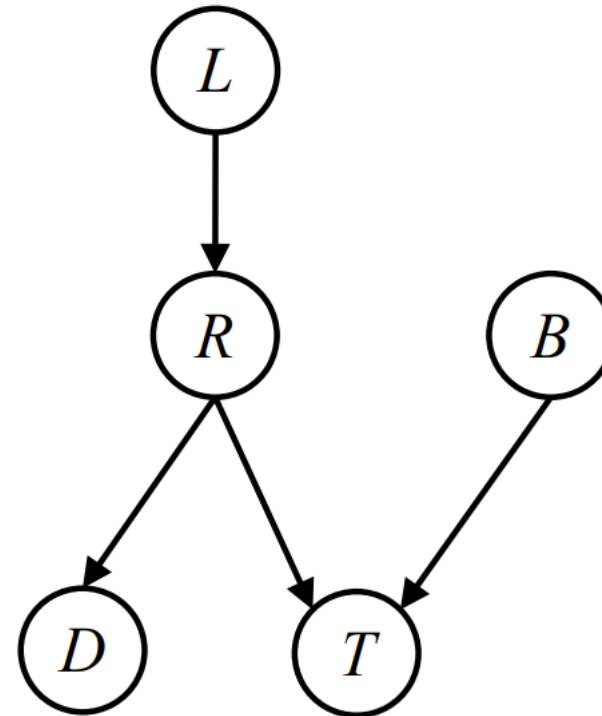


Example: Traffic II

- Let's build a causal graphical model!

- Variables

- T: Traffic
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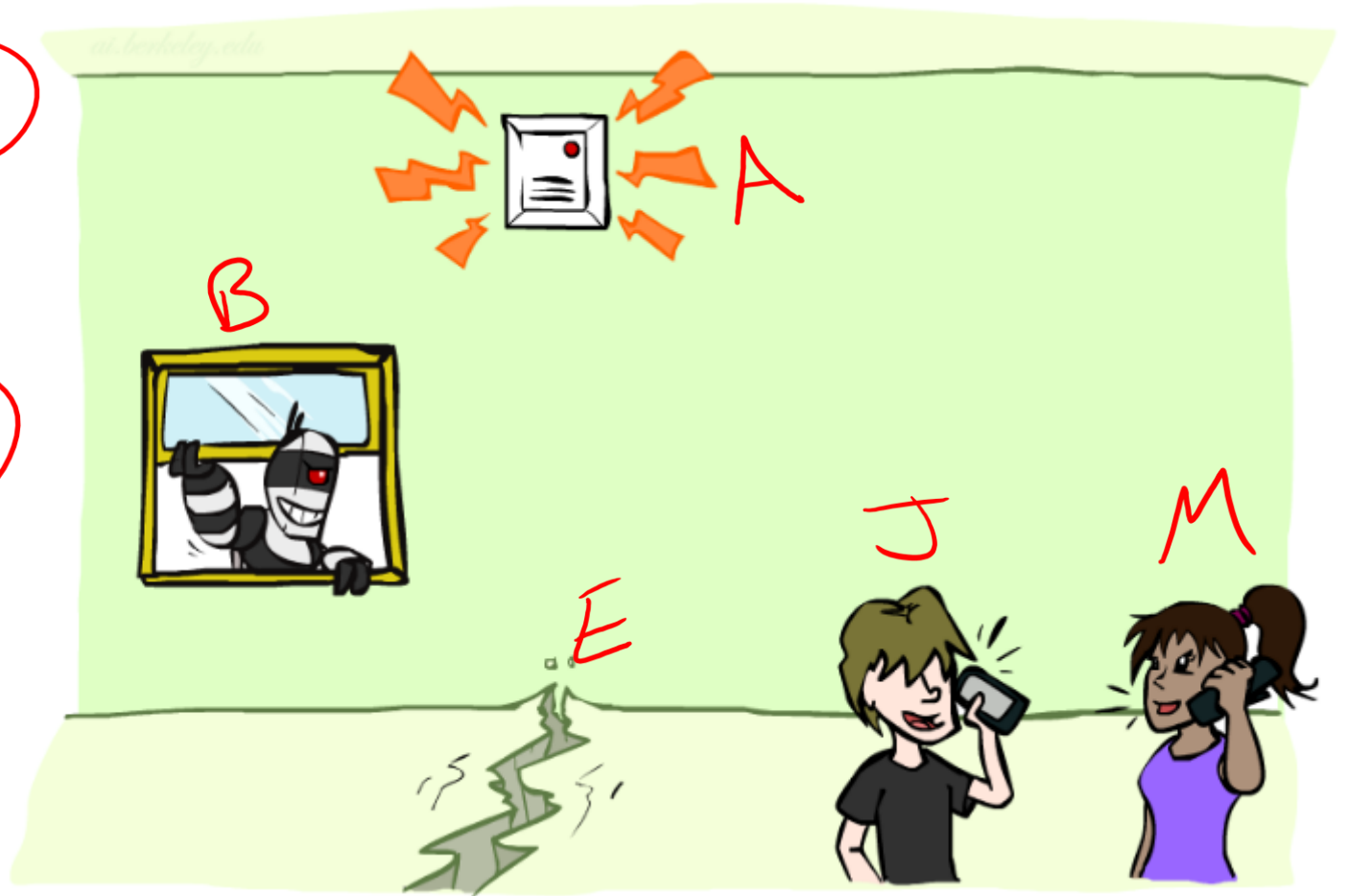
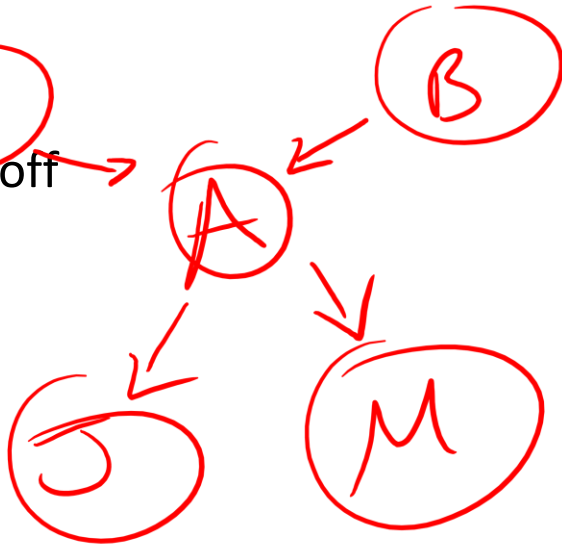


C

Example: Alarm Network

■ Variables

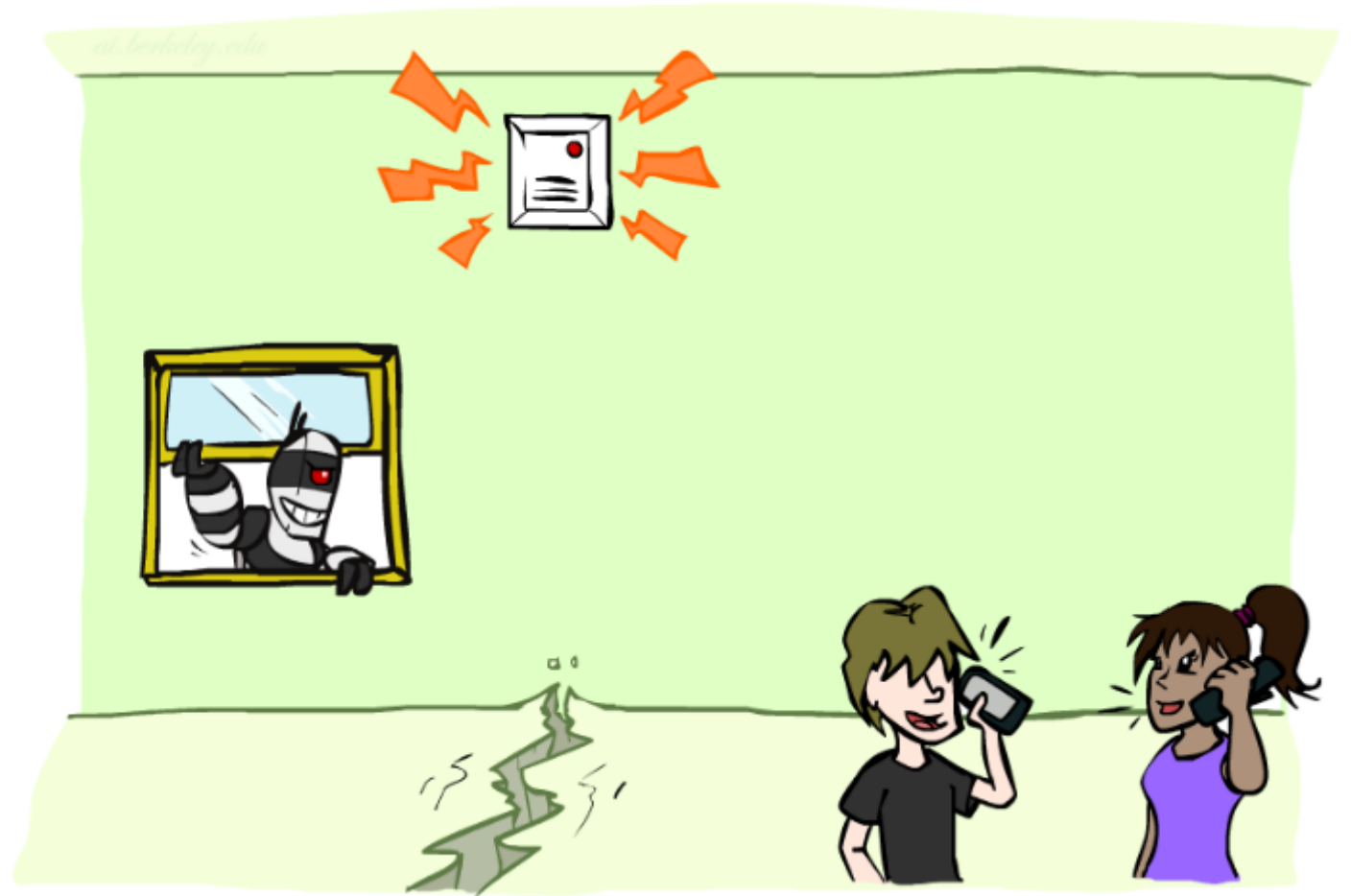
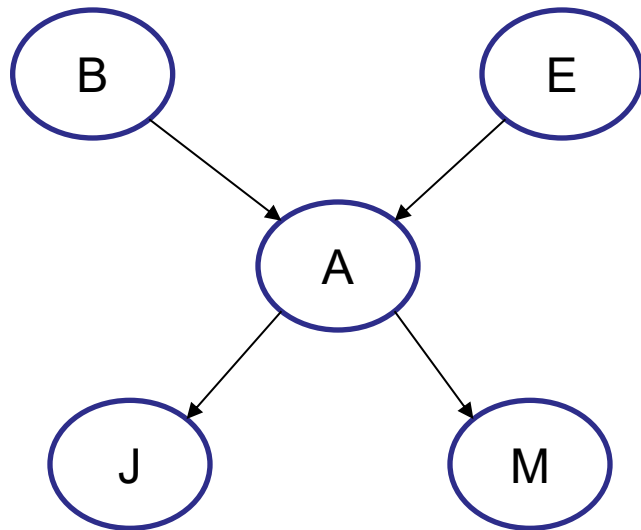
- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Example: Alarm Network

■ Variables

- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Bayes' Net Semantics



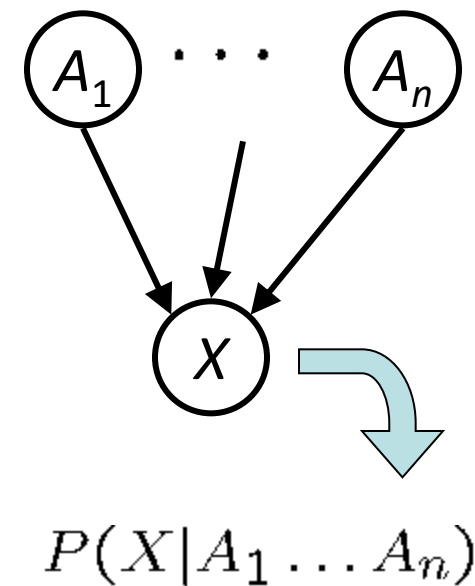
Bayes' Net Semantics



- A set of nodes, one per variable X
- A directed, acyclic graph (DAG)
- A conditional distribution for each node
 - A collection of distributions over X , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

- CPT: conditional probability table



A Bayes net = Topology (graph) + Local Conditional Probabilities

Probabilities in BNs

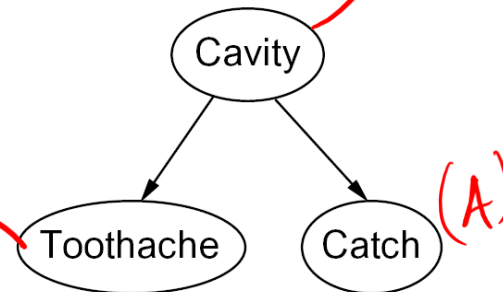
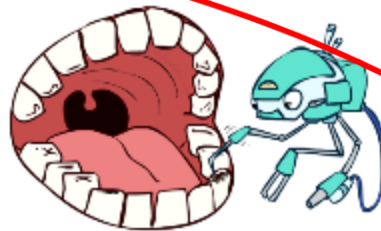


- Bayes' nets **implicitly** encode joint distributions

- As a product of local conditional distributions
- Claim:** To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

- Example:



2^n

$P(T|C)$

+	t	+	c	0.1
+	t	-	c	0.3
-	t	+	c	0.5
-	t	-	c	0.1

$P(C)$

+	c	0.3
-	c	0.7

$$P(C, T, A) = P(C)P(T|C)P(A|C)$$

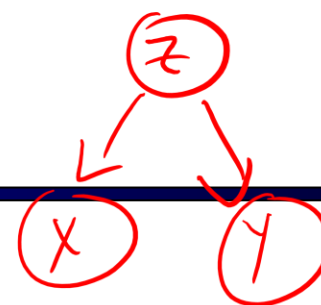
$$P(+cavity, +catch, -toothache) = P(+cavity)P(+catch|+cavity)P(-toothache|+catch)$$

cavity

$$X \perp\!\!\!\perp Y \mid Z$$

$$P(X|Y, Z) = P(X|Z)$$

Probabilities in BNs



- Why are we guaranteed that setting

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

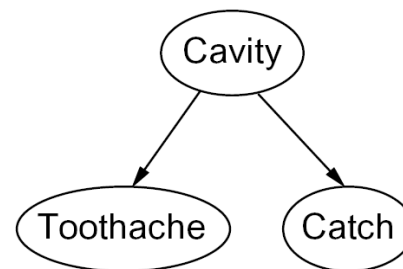
results in a proper joint distribution?

- Chain rule (valid for all distributions): $P(x_1, x_2, \dots, x_n) = \prod_i P(x_i | \underline{x_1 \dots x_{i-1}})$

- Assume conditional independences: $P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(X_i))$

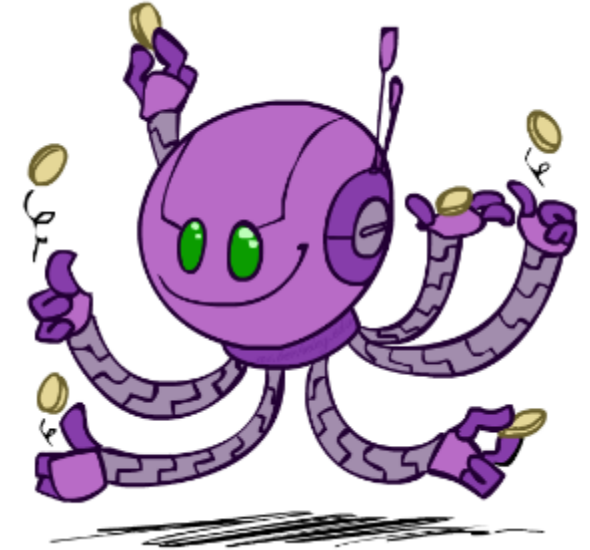
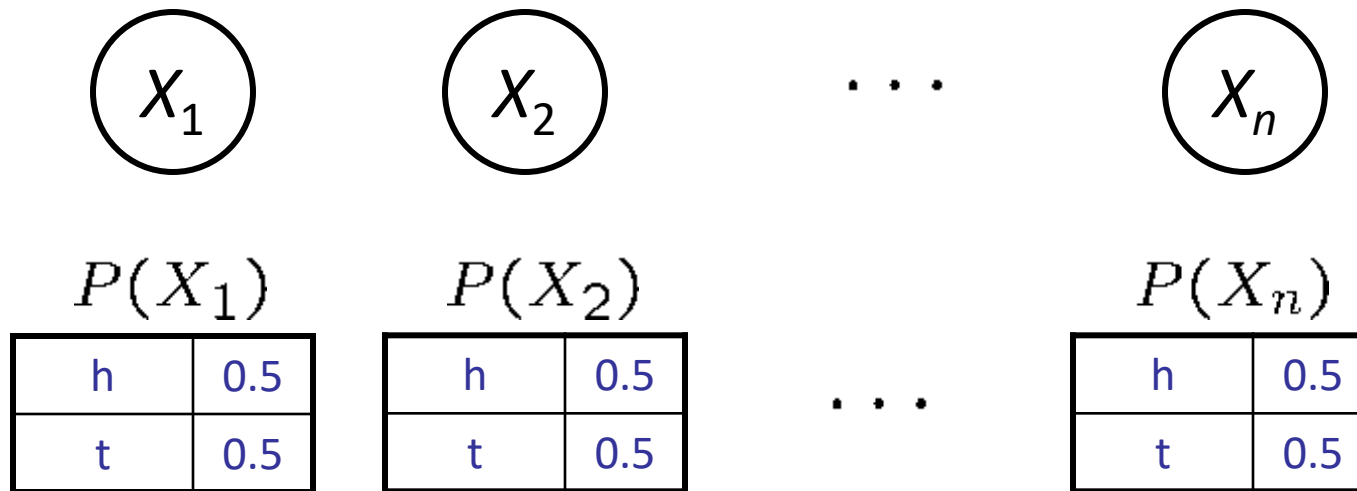
→ Consequence: $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$

- Doesn't say anything about causality (more later)!
- Not every BN can represent every joint distribution



- The topology enforces certain conditional independencies

Example: Coin Flips

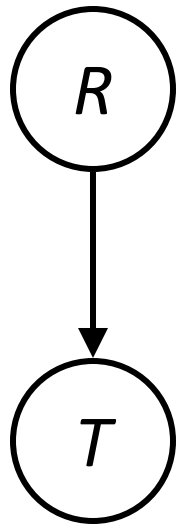


Handwritten red text: $P(X_1=h) P(X_2=h) \dots$

$$P(h, h, t, h) = 0.5 * 0.5 * 0.5 * 0.5$$

Only distributions whose variables are absolutely independent can be represented by a Bayes' net with no arcs.

Example: Traffic



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

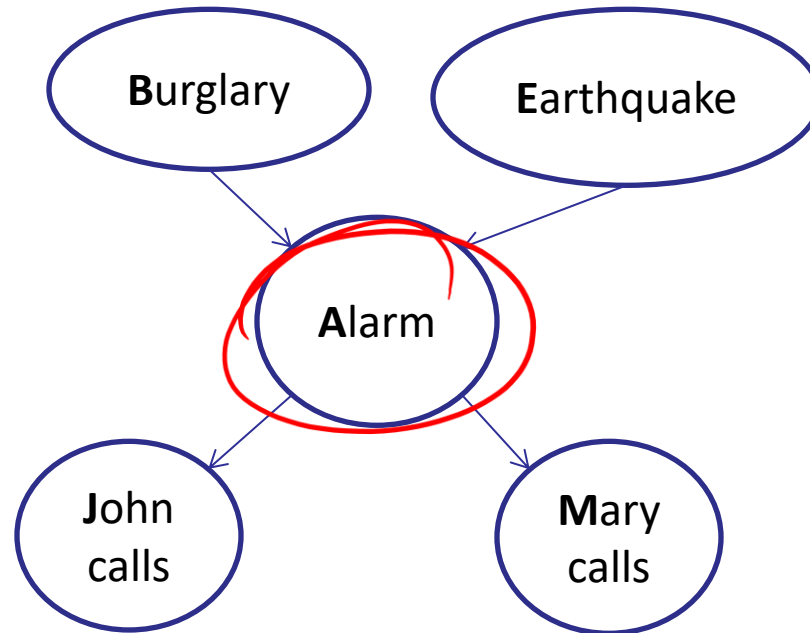
+r	+t	3/4
+r	-t	1/4
-r	+t	1/2
-r	-t	1/2

$$P(+r, -t) = P(R=+r) P(T=-t | R=+r)$$

$\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$



Example: Alarm Network



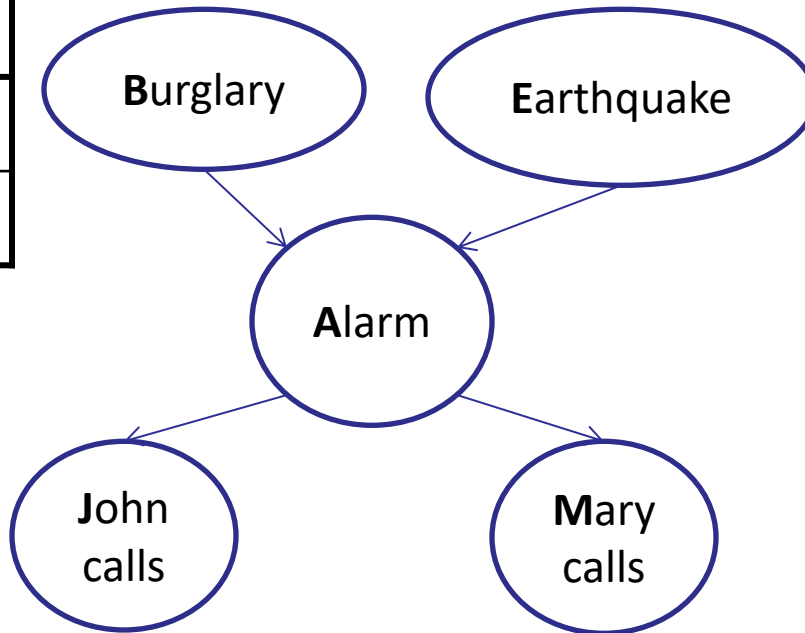
What conditional independence assumptions have we made by drawing this Bayes' Net?

$E \perp\!\!\!\perp B$
 $J \perp\!\!\!\perp M \mid A$
 $J \perp\!\!\!\perp B \mid A$
 $M \perp\!\!\!\perp E \mid A$



Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

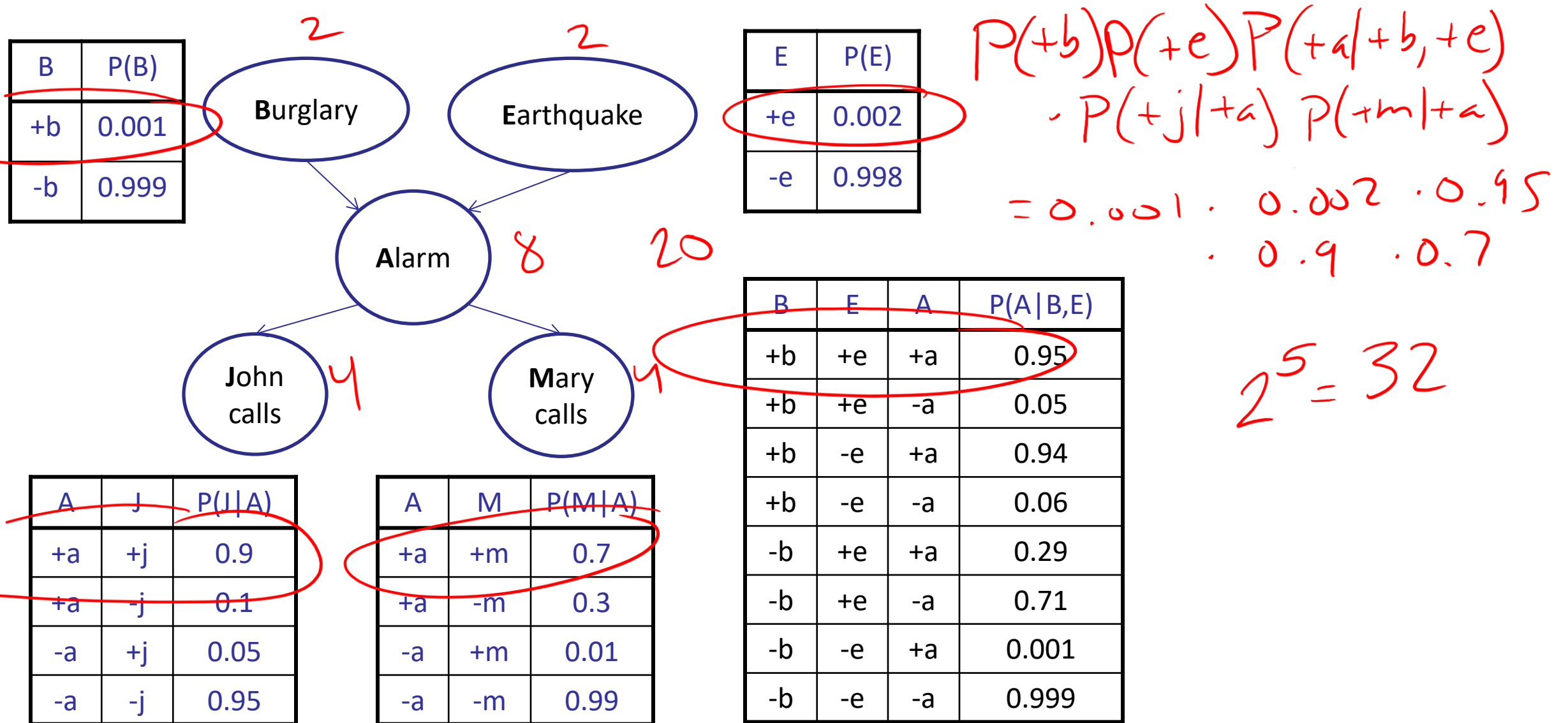


A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

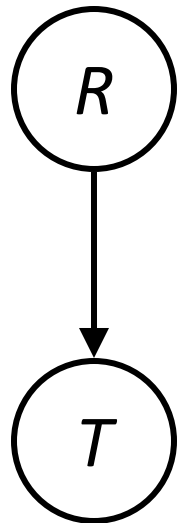
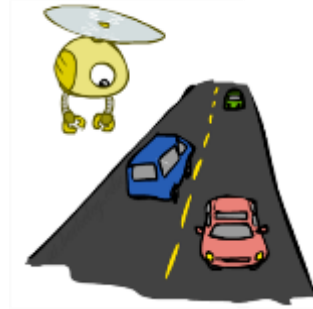
Quiz: $P(+a, +b, +e, +j, +m) = ?$



Example: Traffic

- Causal direction

$$P(+r, +t) = P(+r) P(+t|+r) = \frac{1}{4} \cdot \frac{3}{4}$$



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4

-r	+t	1/2
	-t	1/2

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

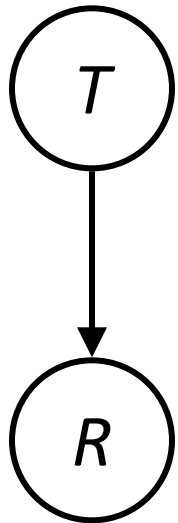
Example: Reverse Traffic

- Reverse causality?

$$P(+t, +r) = P(+t) P(+r | +t)$$

$$= \frac{9}{16} \cdot \frac{1}{3}$$

$$= \frac{3}{16}$$



$P(T)$	
+t	9/16
-t	7/16

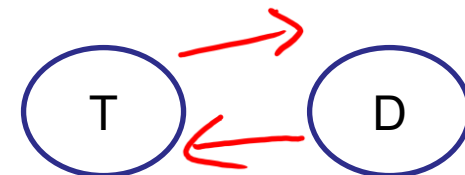
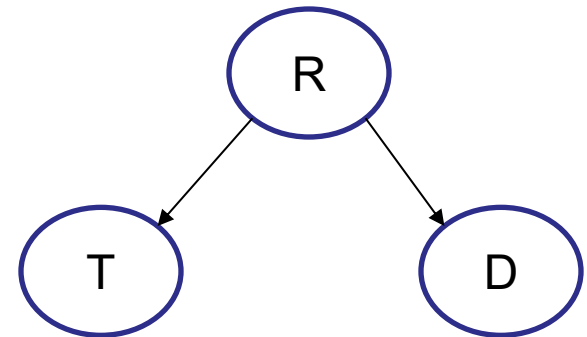
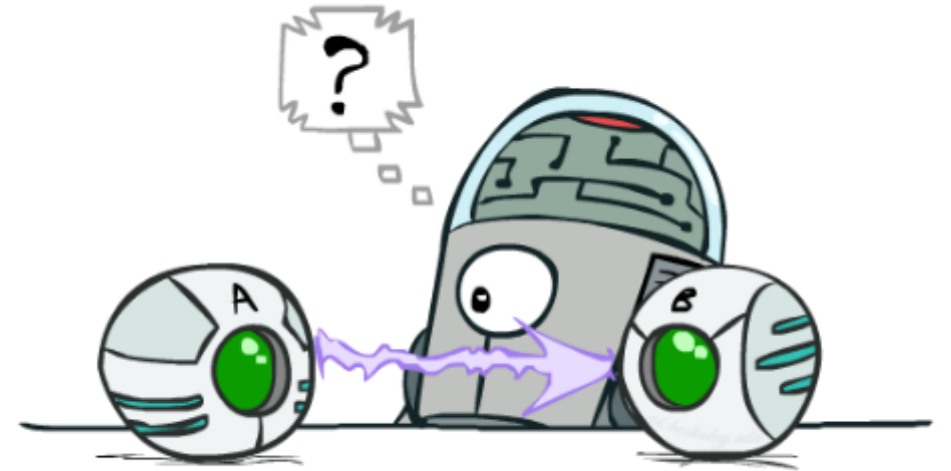
$P(R T)$		
+t	+r	1/3
	-r	2/3
-t	+r	1/7
	-r	6/7

$P(T, R)$

+r	+t	3/16
+r	-t	1/16
-r	+t	6/16
-r	-t	6/16

Causality?

- When Bayes' nets reflect the true causal patterns:
 - Often simpler (nodes have fewer parents)
 - Often easier to think about
 - Often easier to elicit from experts
- BNs need not actually be causal
 - Sometimes no causal net exists over the domain (especially if variables are missing)
 - E.g. consider the variables *Traffic* and *Drips* (and *Rain*)
 - End up with arrows that reflect correlation, not causation
- What do the arrows really mean?
 - Topology may happen to encode causal structure
 - **Topology really encodes conditional independence**
$$P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(X_i))$$



Bayes' Nets

- So far: how a Bayes' net encodes a joint distribution
- Next: how to answer queries about that distribution
 - Today:
 - First assembled BNs using an intuitive notion of conditional independence as causality
 - Then saw that key property is conditional independence
 - Main goal: answer queries about conditional independence and influence
- After that: how to answer numerical queries (inference)

